RAW CUSTOMER DATA (Original — Before

Customer_ID	Name	Age	Gender	City
CUST1000	sheena bonilla	55	Male	Donaldton
CUST1001	Eugene Nelson	26	Female	Phillipsmouth
CUST1002	dr. Benjamin Price	29	Female	Kylebury
CUST1003	jeremy harris	59	Female	Thomastown
CUST1004	Caroline Collins	64	Female	Nicholasshire
CUST1005	Christine Vaughn	24	Male	East Amandamouth
CUST1006	Amber Jackson	29	Female	Deborahborough
CUST1007	Daniel Boyd	29	Female	West Pamelafurt
CUST1008	Melissa Howard	53	Female	Maynardton
CUST1009	Ms. Maria Leach	64	Male	Melissashire
CUST1010	Dominic Page	53	Male	Port Michaelburgh
CUST1011	Brian Lopez	28	Female	East Krista
CUST1012	Kenneth Lambert	55	Female	North Kathryn
CUST1013	Nathan Hernandez	32	Female	New Becky
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth
CUST1015	Laura Alvarez	49	Female	Lawsonhaven
CUST1016	John Scott	24	Female	South Christopher
CUST1017	Adam Cruz	60	Female	Brandonville
CUST1018	Rhonda Moss	43	Male	South William
CUST1019	Michael Torres	46	Female	Gonzalezfort
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh
CUST1021	Jeff Martinez	36	Female	Smithburgh
CUST1022	paul fox	33	Female	Michaelton
CUST1023	Jonathan Steele	52	Female	Molinabury
CUST1024	Heather Collins	40	Male	Rogerschester
CUST1025	Diana Taylor	59	Female	Vasqueztown
CUST1026	Amber Carlson	18	Female	East Michellemouth
CUST1027	Leslie Johnson	31	Male	Mollyshire
CUST1028	Joshua Ryan	27	Female	New David
CUST1029	David Arnold	54	Female	East Jennifer
CUST1030	Jessica Acevedo	64	Female	Amybury
CUST1031	Sarah Vance	41	Male	Johnsonville
CUST1032	Mike Jennings	48	Female	Fostermouth
CUST1033	Robert Jones DDS	25	Male	Tristanton
CUST1034	Christopher Olson	27	Female	West Richard
CUST1035	Rachel Myers	52	Male	Robertberg
CUST1036	Thomas Phillips	20	Female	Luisfurt
CUST1037	Christopher Carter	27	Female	Brendastad
CUST1038	Deanna Simpson	35	Female	Brownport
CUST1039	Tonya Reynolds	55	Female	Darleneton

CUST1040	Eric Vega	50	Male	Williamsonland
CUST1041	James Gibbs	21	Male	Mooneyshire
CUST1042	William Martin	19	Male	Ryanburgh
CUST1043	Joseph Snow	45	Male	South Michaelfort
CUST1044	Denise Williams	34	Male	Christianstad
CUST1045	Jordan Jackson DDS	54	Male	North Johnview
CUST1046	Michelle Garcia DDS	55	Female	Perezbury
CUST1047	Gina Knight	54	Female	Lake Roberthaven
CUST1048	April Mullins	54	Male	Dawsonbury
CUST1049	Timothy Weaver	45	Female	New Amy
CUST1050	Matthew Hunter Jr.	61	Female	Gregoryview

⚠ Issues found (highlighted in red):

CUST1000 Missing Loyalty Level; Name has extra spaces
CUST1002 Title prefix in name (dr.); wrong capitalisation
CUST1003 Name not title-cased (jeremy harris)
CUST1009 Title prefix (Ms.) in name
CUST1018 Missing Loyalty Level
CUST1022 Name has leading spaces (paul fox)
CUST1027 Missing Loyalty Level
CUST1033 Professional suffix in name (DDS)
CUST1037 Missing Loyalty Level

Cleaning)

State	Country	Loyalty_Level
North Carolina	USA	
North Dakota	USA	Platinum
Alabama	USA	Platinum
Ohio	USA	Bronze
California	USA	Silver
Oregon	USA	Bronze
South Dakota	USA	Platinum
West Virginia	USA	Bronze
New Hampshire	USA	Gold
Pennsylvania	USA	Platinum
Pennsylvania	USA	Bronze
New York	USA	Gold
Kansas	USA	Platinum
Montana	USA	Silver
Nebraska	USA	Silver
Washington	USA	Platinum
Pennsylvania	USA	Bronze
Georgia	USA	Platinum
Kansas	USA	
New Jersey	USA	Platinum
Delaware	USA	Silver
Oklahoma	USA	Gold
Hawaii	USA	Silver
Oregon	USA	Silver
Idaho	USA	Platinum
California	USA	Bronze
Missouri	USA	Gold
Utah	USA	
New Jersey	USA	Platinum
California	USA	Silver
Pennsylvania	USA	Bronze
Oregon	USA	Silver
Arizona	USA	Gold
Oklahoma	USA	Platinum
Maryland	USA	Platinum
North Dakota	USA	Silver
Georgia	USA	Gold
South Dakota	USA	
North Dakota	USA	Bronze
Michigan	USA	Gold

Kansas	USA	Gold
Alaska	USA	Platinum
Connecticut	USA	Platinum
Vermont	USA	Gold
Hawaii	USA	Bronze
Arizona	USA	Gold
Ohio	USA	Platinum
Nevada	USA	Silver
North Carolina	USA	Platinum
Ohio	USA	Gold
Illinois	USA	Gold

CLEANED CUSTOMER DA

Customer_ID	Name (Cleaned)	Age	Gender	City (Cleaned)
CUST1000	Sheena Bonilla	55	Male	Donaldton
CUST1001	Eugene Nelson	26	Female	Phillipsmouth
CUST1002	Benjamin Price	29	Female	Kylebury
CUST1003	Jeremy Harris	59	Female	Thomastown
CUST1004	Caroline Collins	64	Female	Nicholasshire
CUST1005	Christine Vaughn	24	Male	East Amandamouth
CUST1006	Amber Jackson	29	Female	Deborahborough
CUST1007	Daniel Boyd	29	Female	West Pamelafurt
CUST1008	Melissa Howard	53	Female	Maynardton
CUST1009	Maria Leach	64	Male	Melissashire
CUST1010	Dominic Page	53	Male	Port Michaelburgh
CUST1011	Brian Lopez	28	Female	East Krista
CUST1012	Kenneth Lambert	55	Female	North Kathryn
CUST1013	Nathan Hernandez	32	Female	New Becky
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth
CUST1015	Laura Alvarez	49	Female	Lawsonhaven
CUST1016	John Scott	24	Female	South Christopher
CUST1017	Adam Cruz	60	Female	Brandonville
CUST1018	Rhonda Moss	43	Male	South William
CUST1019	Michael Torres	46	Female	Gonzalezfort
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh
CUST1021	Jeff Martinez	36	Female	Smithburgh
CUST1022	Paul Fox	33	Female	Michaelton
CUST1023	Jonathan Steele	52	Female	Molinabury
CUST1024	Heather Collins	40	Male	Rogerschester
CUST1025	Diana Taylor	59	Female	Vasqueztown
CUST1026	Amber Carlson	18	Female	East Michellemouth
CUST1027	Leslie Johnson	31	Male	Mollyshire
CUST1028	Joshua Ryan	27	Female	New David
CUST1029	David Arnold	54	Female	East Jennifer
CUST1030	Jessica Acevedo	64	Female	Amybury
CUST1031	Sarah Vance	41	Male	Johnsonville
CUST1032	Mike Jennings	48	Female	Fostermouth
CUST1033	Robert Jones	25	Male	Tristanton
CUST1034	Christopher Olson	27	Female	West Richard
CUST1035	Rachel Myers	52	Male	Robertberg
CUST1036	Thomas Phillips	20	Female	Luisfurt
CUST1037	Christopher Carter	27	Female	Brendastad
CUST1038	Deanna Simpson	35	Female	Brownport
CUST1039	Tonya Reynolds	55	Female	Darleneton

CUST1040	Eric Vega	50	Male	Williamsonland
CUST1041	James Gibbs	21	Male	Mooneyshire
CUST1042	William Martin	19	Male	Ryanburgh
CUST1043	Joseph Snow	45	Male	South Michaelfort
CUST1044	Denise Williams	34	Male	Christianstad
CUST1045	Jordan Jackson	54	Male	North Johnview
CUST1046	Michelle Garcia	55	Female	Perezbury
CUST1047	Gina Knight	54	Female	Lake Roberthaven
CUST1048	April Mullins	54	Male	Dawsonbury
CUST1049	Timothy Weaver	45	Female	New Amy
CUST1050	Matthew Hunter	61	Female	Gregoryview

ATA

State	Country (Std)	Loyalty_Level	Age_Group
North Carolina	USA	Bronze	55-64
North Dakota	USA	Platinum	25-34
Alabama	USA	Platinum	25-34
Ohio	USA	Bronze	55-64
California	USA	Silver	55-64
Oregon	USA	Bronze	18-24
South Dakota	USA	Platinum	25-34
West Virginia	USA	Bronze	25-34
New Hampshire	USA	Gold	45-54
Pennsylvania	USA	Platinum	55-64
Pennsylvania	USA	Bronze	45-54
New York	USA	Gold	25-34
Kansas	USA	Platinum	55-64
Montana	USA	Silver	25-34
Nebraska	USA	Silver	45-54
Washington	USA	Platinum	45-54
Pennsylvania	USA	Bronze	18-24
Georgia	USA	Platinum	55-64
Kansas	USA	Bronze	35-44
New Jersey	USA	Platinum	45-54
Delaware	USA	Silver	55-64
Oklahoma	USA	Gold	35-44
Hawaii	USA	Silver	25-34
Oregon	USA	Silver	45-54
Idaho	USA	Platinum	35-44
California	USA	Bronze	55-64
Missouri	USA	Gold	18-24
Utah	USA	Bronze	25-34
New Jersey	USA	Platinum	25-34
California	USA	Silver	45-54
Pennsylvania	USA	Bronze	55-64
Oregon	USA	Silver	35-44
Arizona	USA	Gold	45-54
Oklahoma	USA	Platinum	25-34
Maryland	USA	Platinum	25-34
North Dakota	USA	Silver	45-54
Georgia	USA	Gold	18-24
South Dakota	USA	Bronze	25-34
North Dakota	USA	Bronze	35-44
Michigan	USA	Gold	55-64

Kansas	USA	Gold	45-54
Alaska	USA	Platinum	18-24
Connecticut	USA	Platinum	18-24
Vermont	USA	Gold	45-54
Hawaii	USA	Bronze	25-34
Arizona	USA	Gold	45-54
Ohio	USA	Platinum	55-64
Nevada	USA	Silver	45-54
North Carolina	USA	Platinum	45-54
Ohio	USA	Gold	45-54
Illinois	USA	Gold	55-64

RAW PRODUCT DATA (Original — E

Product_ID	Product_Name	Category	Sub_Category
PROD0	Brother Laser Printer	Electronics	Computer
PROD1	Men's Graphic T-Shirt	Clothing	Tops
PROD2	Women's Casual Hoodie	Clothing	Outerwear
P-3	Men's Running Jacket	Clothing	Outerwear
PROD4	Women's Yoga Pants	Clothing	Bottoms
PROD5	Men's Leather Belt	Clothing	Accessories
PROD6	Herbal Face Moisturizer	Beauty	Skincare
PROD7	LED Table Lamp	Home	Lighting
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio
PROD9	Adjustable Dumbbell Set	Sports	Fitness
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen
PROD11	HD Smart TV Stick	Electronics	Accessories
PROD12	Hair Strengthening Shampoo	Beauty	Haircare
PROD13	USB-C Fast Charger	Electronics	Accessories
PROD14	Decorative Wall Mirror	Home	Decor
PROD15	4K Home Projector	Electronics	Appliances
P-16	Beard Grooming Kit	Beauty	Grooming
PROD17	Soccer Training Cone Set	Sports	Accessories
PROD18	Women's Denim Jeans	Clothing	Bottoms
PROD19	Knee Support Brace	Sports	Equipment
PROD20	Outdoor Sports Water Bottle	Sports	Accessories
PROD21	Multi-Purpose Storage Box	Home	Storage
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment
PROD23	Men's Formal Shirt	Clothing	Tops
PROD24	Foldable Camping Chair	Sports	Outdoor
PROD25	Women's Scarf Set	Clothing	Accessories
PROD26	Wireless Bluetooth Speaker	Electronics	Audio
PROD27	Decorative Floor Vase	Home	Decor
PROD28	Sports Stopwatch Timer	Sports	Accessories
PROD29	Fabric Sofa Throw Blanket	Home	Decor
PROD30	Stainless Steel Cutlery Set	Home	Kitchen
PROD31	Basketball	Sports	Team Sports
PROD32	Vitamin C Face Serum	Beauty	Skincare
PROD33	Modern Wall Clock	Home	Decor
PROD34	Gaming Mechanical Keyboard	Electronics	Computer
PROD35	Bedroom Table Lamp	Home	Lighting
PROD36	Women's Summer Dress	Clothing	Tops
PROD37	Men's Casual Sneakers	Clothing	Footwear
PROD38	Sports Water Hydration Pack	Sports	Outdoor
PROD39	Portable Power Bank	Electronics	Mobile

PROD40	Noise-Canceling Headphones	Electronics	Audio
PROD41	Yoga Exercise Mat	Sports	Fitness
PROD42	Smart Home WiFi Camera	Electronics	Accessories
PROD43	USB External Hard Drive	Electronics	Computer
PROD44	Adjustable Jump Rope	Sports	Fitness
PROD45	High-Grip Football	Sports	Team Sports
PROD46	Men's Winter Jacket	Clothing	Outerwear
PROD47	Makeup Brush Set	Beauty	Makeup
PROD48	Car Dashboard Phone Holder	Electronics	Accessories
PROD49	Men's Roll-On Perfume	Beauty	Fragrance
PROD50	Boxing Hand Wraps	Sports	Fitness

Before Cleaning)

Brand	Cost	Stock
Hunt Inc	112.42	416
Terry-Moore	342.56	181
Sims Inc	MISSING	352
Pineda Ltd	364.67	452
Gonzales-Davis	306.46	487
Baldwin, Garza and Gonzalez	261	59
Davidson PLC	312.8	350
Harrell LLC	218.86	170
Perez-Johnston	439.07	142
Hill PLC	495.72	323
Hamilton Group	298.86	308
Gibson, Hobbs and Weaver	449.4	MISSING
Rice LLC	115.82	460
Dawson-Perkins	242.23	453
Luna, Anderson and Bowman	48.01	439
Thomas LLC	465.34	301
Erickson-Green	337.42	256
Ford, Murphy and Wilson	144.53	367
Guerra-Hendrix	MISSING	343
May-Reese	21.34	65
Soto-Ellis	131.51	MISSING
Gardner-Montes	123.46	76
Flores, Pineda and Rodriguez	484.62	432
Moon, Mora and Santos	MISSING	360
Cook Ltd	166.06	340
Brown and Sons	174.02	268
Reed-Schmidt	322.41	213
Khan, Jones and Paul	MISSING	378
Burns-Hansen	38.76	137
Hoffman Group	218.48	276
Stevens, Fernandez and Martin	19.52	57
Nolan-Snow	204.39	281
Griffin Ltd	283.83	244
Weiss-Watson	179.77	305
Lucero LLC	419.28	86
Jones LLC	382.22	329
Smith-Mills	223.43	154
Hogan-Lee	353.47	214
Sims-Smith	135.61	394
Cummings and Sons	326.06	301

Taylor-Hart	38.52	378
Carroll, Long and Mendoza	313.26	369
Holmes LLC	450.71	389
Robinson Inc	253.33	266
Cox Inc	167.12	316
Bridges, Banks and Cummings	318.91	114
Kelly, Smith and Perez	167.34	296
Cook Inc	MISSING	431
Byrd, Stewart and Valdez	177.98	103
Sullivan and Sons	43.29	267
Smith-King	190.29	260

CLEANED PRODU

Product_ID	Product_Name	Category	Sub_Category
PROD0	Brother Laser Printer	Electronics	Computer
PROD1	Men's Graphic T-Shirt	Clothing	Tops
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PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen
PROD11	HD Smart TV Stick	Electronics	Accessories
PROD12	Hair Strengthening Shampoo	Beauty	Haircare
PROD13	USB-C Fast Charger	Electronics	Accessories
PROD14	Decorative Wall Mirror	Home	Decor
PROD15	4K Home Projector	Electronics	Appliances
PROD16	Beard Grooming Kit	Beauty	Grooming
PROD17	Soccer Training Cone Set	Sports	Accessories
PROD18	Women's Denim Jeans	Clothing	Bottoms
PROD19	Knee Support Brace	Sports	Equipment
PROD20	Outdoor Sports Water Bottle	Sports	Accessories
PROD21	Multi-Purpose Storage Box	Home	Storage
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment
PROD23	Men's Formal Shirt	Clothing	Tops
PROD24	Foldable Camping Chair	Sports	Outdoor
PROD25	Women's Scarf Set	Clothing	Accessories
PROD26	Wireless Bluetooth Speaker	Electronics	Audio
PROD27	Decorative Floor Vase	Home	Decor
PROD28	Sports Stopwatch Timer	Sports	Accessories
PROD29	Fabric Sofa Throw Blanket	Home	Decor
PROD30	Stainless Steel Cutlery Set	Home	Kitchen
PROD31	Basketball	Sports	Team Sports
PROD32	Vitamin C Face Serum	Beauty	Skincare
PROD33	Modern Wall Clock	Home	Decor
PROD34	Gaming Mechanical Keyboard	Electronics	Computer
PROD35	Bedroom Table Lamp	Home	Lighting
PROD36	Women's Summer Dress	Clothing	Tops
PROD37	Men's Casual Sneakers	Clothing	Footwear
PROD38	Sports Water Hydration Pack	Sports	Outdoor
PROD39	Portable Power Bank	Electronics	Mobile

PROD40	Noise-Canceling Headphones	Electronics	Audio
PROD41	Yoga Exercise Mat	Sports	Fitness
PROD42	Smart Home WiFi Camera	Electronics	Accessories
PROD43	USB External Hard Drive	Electronics	Computer
PROD44	Adjustable Jump Rope	Sports	Fitness
PROD45	High-Grip Football	Sports	Team Sports
PROD46	Men's Winter Jacket	Clothing	Outerwear
PROD47	Makeup Brush Set	Beauty	Makeup
PROD48	Car Dashboard Phone Holder	Electronics	Accessories
PROD49	Men's Roll-On Perfume	Beauty	Fragrance
PROD50	Boxing Hand Wraps	Sports	Fitness

CT DATA

Brand	Cost	Stock	Stock_Status
Hunt Inc	112.42	416	High
Terry-Moore	342.56	181	Medium
Sims Inc	280	352	High
Pineda Ltd	364.67	452	High
Gonzales-Davis	306.46	487	High
Baldwin, Garza and Gonzalez	261	59	Low
Davidson PLC	312.8	350	High
Harrell LLC	218.86	170	Medium
Perez-Johnston	439.07	142	Medium
Hill PLC	495.72	323	High
Hamilton Group	298.86	308	High
Gibson, Hobbs and Weaver	449.4	280	High
Rice LLC	115.82	460	High
Dawson-Perkins	242.23	453	High
Luna, Anderson and Bowman	48.01	439	High
Thomas LLC	465.34	301	High
Erickson-Green	337.42	256	High
Ford, Murphy and Wilson	144.53	367	High
Guerra-Hendrix	280	343	High
May-Reese	21.34	65	Low
Soto-Ellis	131.51	280	High
Gardner-Montes	123.46	76	Low
Flores, Pineda and Rodriguez	484.62	432	High
Moon, Mora and Santos	280	360	High
Cook Ltd	166.06	340	High
Brown and Sons	174.02	268	High
Reed-Schmidt	322.41	213	High
Khan, Jones and Paul	220	378	High
Burns-Hansen	38.76	137	Medium
Hoffman Group	218.48	276	High
Stevens, Fernandez and Martin	19.52	57	Low
Nolan-Snow	204.39	281	High
Griffin Ltd	283.83	244	High
Weiss-Watson	179.77	305	High
Lucero LLC	419.28	86	Medium
Jones LLC	382.22	329	High
Smith-Mills	223.43	154	Medium
Hogan-Lee	353.47	214	High
Sims-Smith	135.61	394	High
Cummings and Sons	326.06	301	High

Taylor-Hart	38.52	378	High
Carroll, Long and Mendoza	313.26	369	High
Holmes LLC	450.71	389	High
Robinson Inc	253.33	266	High
Cox Inc	167.12	316	High
Bridges, Banks and Cummings	318.91	114	Medium
Kelly, Smith and Perez	167.34	296	High
Cook Inc	**220**	431	High
Byrd, Stewart and Valdez	177.98	103	Medium
Sullivan and Sons	43.29	267	High
Smith-King	190.29	260	High

STORE DATA

Store_ID	Store_Name	Region	City	Store_Type
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins and Johnson	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones and Park	West	Thompsonborough	Online
STORE9	Moore, Padilla and Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner and Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges and Harris	North	Sabrinatown	Online
STORE13	Gardner, Humphrey and Contreras	East	Johnsonville	Flagship
STORE14	Mcdonald Inc	South	Christopherland	Outlet
STORE15	Shelton PLC	North	East Jenniferstad	Flagship
STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
STORE17	Holloway Ltd	North	Bensonhaven	Flagship
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
STORE19	Mann-Austin	East	Vasquezborough	Online

CLEANED SALES DA

Sale_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount
SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	218.04	9.00%
SALE1	2023-12-25	CUST1467	PROD74	STORE17	3	104.47	27.00%
SALE2	2024-06-03	CUST1056	PROD49	STORE8	4	914.05	25.00%
SALE3	2023-11-29	CUST1346	PROD76	STORE8	1	921.51	16.00%
SALE4	2024-01-06	CUST1493	PROD59	STORE5	3	745.30	18.00%
SALE5	2024-07-26	CUST1116	PROD15	STORE7	3	147.01	19.00%
SALE6	2025-07-12	CUST1016	PROD56	STORE4	2	785.31	15.00%
SALE7	2023-11-18	CUST1152	PROD70	STORE8	4	342.39	23.00%
SALE8	2025-05-04	CUST1187	PROD99	STORE5	1	448.25	7.00%
SALE9	2025-07-31	CUST1464	PROD33	STORE0	2	873.17	23.00%
SALE10	2024-02-13	CUST1065	PROD24	STORE14	1	278.63	7.00%
SALE11	2025-05-24	CUST1320	PROD39	STORE1	4	286.92	17.00%
SALE12	2024-08-24	CUST1237	PROD88	STORE4	4	644.58	1.00%
SALE13	2025-08-06	CUST1443	PROD34	STORE16	2	316.90	30.00%
SALE14	2025-06-21	CUST1348	PROD24	STORE15	4	403.47	29.00%
SALE15	2024-12-10	CUST1435	PROD36	STORE9	1	871.34	4.00%
SALE16	2025-09-13	CUST1276	PROD65	STORE19	2	451.98	25.00%
SALE17	2024-12-11	CUST1079	PROD27	STORE4	2	502.69	23.00%
SALE18	2024-08-13	CUST1094	PROD39	STORE4	2	559.23	25.00%
SALE19	2024-03-18	CUST1239	PROD60	STORE13	4	0.00	4.00%
SALE20	2025-07-05	CUST1144	PROD72	STORE17	3	986.04	27.00%
SALE21	2024-06-04	CUST1396	PROD43	STORE18	1	589.56	22.00%
SALE22	2024-04-12	CUST1476	PROD69	STORE10	3	979.42	3.00%
SALE23	2024-06-23	CUST1368	PROD12	STORE8	1	719.28	8.00%
SALE24	2025-09-08	CUST1186	PROD67	STORE6	3	77.60	7.00%
SALE25	2024-09-17	CUST1381	PROD5	STORE3	4	852.71	14.00%
SALE26	2024-01-03	CUST1213	PROD98	STORE8	4	156.16	0.00%
SALE27	2025-02-22	CUST1200	PROD77	STORE8	3	815.49	17.00%
SALE28	2025-09-23	CUST1148	PROD13	STORE2	2	74.21	11.00%
SALE29	2024-01-11	CUST1348	PROD99	STORE3	1	451.67	15.00%
SALE30	2023-10-18	CUST1028	PROD5	STORE19	1	429.38	13.00%
SALE31	2024-10-28	CUST1412	PROD53	STORE12	2	148.96	3.00%
SALE32	2024-11-01	CUST1398	PROD75	STORE8	4	313.83	8.00%
SALE33	2024-11-24	CUST1453	PROD12	STORE6	2	717.31	15.00%
SALE34	2023-10-16	CUST1344	PROD79	STORE1	1	873.22	18.00%
SALE35	2025-08-09	CUST1218	PROD16	STORE3	4	109.36	19.00%
SALE36	2024-12-09	CUST1086	PROD5	STORE7	3	559.52	16.00%
SALE37	2024-02-07	CUST1492	PROD10	STORE17	3	121.76	27.00%
SALE38	2024-03-27	CUST1189	PROD33	STORE13	4	735.32	6.00%
SALE39	2023-12-04	CUST1431	PROD94	STORE3	2	600.73	8.00%

SALE40	2024-07-13	CUST1072	PROD26	STORE19	2	873.86	7.00%
SALE41	2023-11-11	CUST1302	PROD23	STORE8	3	342.47	8.00%
SALE42	2024-12-23	CUST1335	PROD29	STORE9	2	915.15	16.00%
SALE43	2025-01-20	CUST1084	PROD33	STORE17	1	863.11	28.00%
SALE44	2024-09-03	CUST1203	PROD43	STORE12	2	209.32	23.00%
SALE45	2024-11-10	CUST1470	PROD59	STORE8	4	288.60	9.00%
SALE46	2024-10-05	CUST1070	PROD78	STORE16	1	418.69	6.00%
SALE47	2024-06-23	CUST1008	PROD30	STORE11	1	135.78	14.00%
SALE48	2023-10-19	CUST1203	PROD44	STORE17	1	517.13	28.00%
SALE49	2024-11-11	CUST1420	PROD21	STORE16	2	384.66	8.00%
SALE50	2023-12-31	CUST1070	PROD52	STORE9	3	83.70	24.00%
SALE51	2025-02-01	CUST1142	PROD69	STORE7	3	76.57	26.00%
SALE52	2025-08-25	CUST1148	PROD89	STORE6	3	0.00	0.00%
SALE53	2025-01-20	CUST1324	PROD20	STORE8	1	900.35	13.00%
SALE54	2025-02-18	CUST1095	PROD40	STORE10	4	428.09	7.00%
SALE55	2024-10-24	CUST1416	PROD49	STORE11	2	559.71	30.00%
SALE56	2023-12-22	CUST1080	PROD34	STORE6	2	833.70	6.00%
SALE57	2024-06-19	CUST1467	PROD47	STORE14	2	680.16	28.00%
SALE58	2024-12-07	CUST1404	PROD15	STORE7	4	733.42	0.00%
SALE59	2023-11-27	CUST1367	PROD89	STORE11	1	996.33	23.00%
SALE60	2025-07-16	CUST1408	PROD31	STORE10	1	198.83	9.00%
SALE61	2025-02-28	CUST1239	PROD19	STORE7	1	374.92	18.00%
SALE62	2024-08-11	CUST1464	PROD93	STORE16	1	468.94	12.00%
SALE63	2025-09-27	CUST1126	PROD89	STORE12	2	121.37	5.00%
SALE64	2025-09-14	CUST1255	PROD53	STORE10	3	480.06	1.00%
SALE65	2023-12-30	CUST1162	PROD99	STORE17	2	856.57	24.00%
SALE66	2023-10-14	CUST1269	PROD77	STORE2	4	814.25	18.00%
SALE67	2023-11-25	CUST1483	PROD12	STORE2	1	989.17	20.00%
SALE68	2023-12-10	CUST1372	PROD72	STORE10	4	62.48	18.00%
SALE69	2025-08-28	CUST1423	PROD13	STORE16	4	561.24	24.00%
SALE70	2025-07-11	CUST1058	PROD60	STORE5	3	92.24	28.00%
SALE71	2025-08-20	CUST1168	PROD72	STORE1	2	668.30	7.00%
SALE72	2024-08-29	CUST1409	PROD38	STORE5	1	553.82	9.00%
SALE73	2024-11-30	CUST1037	PROD45	STORE11	2	556.43	13.00%
SALE74	2025-10-12	CUST1273	PROD20	STORE8	1	184.98	10.00%
SALE75	2024-07-27	CUST1035	PROD97	STORE12	4	700.18	23.00%
SALE76	2023-12-25	CUST1019	PROD76	STORE10	2	686.31	25.00%
SALE77	2025-05-23	CUST1007	PROD67	STORE15	1	687.82	27.00%
SALE78	2024-11-13	CUST1221	PROD83	STORE8	2	900.55	16.00%
SALE79	2023-12-08	CUST1115	PROD21	STORE2	1	341.17	15.00%
SALE80	2025-08-16	CUST1176	PROD52	STORE8	1	442.27	6.00%

Legend:

Imputed Quantity (was missing → set to 1)

	Imputed Price/Total (was missing → calculated)
	XLOOKUP-enriched columns (from related tables)

TA — WITH XLOOKUP ENRICHMENT

Payment_Type	Total_Amount	Customer_Name	Product_Name
Credit Card	198.42	Unknown	Unknown
PayPal	228.79	Unknown	Unknown
Credit Card	2,742.15	Unknown	Men's Roll-On Perfume
COD	774.07	Unknown	Unknown
Credit Card	1,833.44	Unknown	Unknown
COD	357.23	Unknown	4K Home Projector
PayPal	1,335.03	John Scott	Unknown
PayPal	1,054.56	Unknown	Unknown
PayPal	1,250.62	Unknown	Unknown
PayPal	1,344.68	Unknown	Modern Wall Clock
Credit Card	259.13	Unknown	Foldable Camping Chair
Credit Card	952.57	Unknown	Portable Power Bank
Credit Card	2,552.54	Unknown	Unknown
PayPal	443.66	Unknown	Gaming Mechanical Keyboard
PayPal	1,145.85	Unknown	Foldable Camping Chair
COD	2,509.46	Unknown	Women's Summer Dress
Credit Card	677.97	Unknown	Unknown
Credit Card	774.14	Unknown	Decorative Floor Vase
COD	838.85	Unknown	Portable Power Bank
COD	1,782.45	Unknown	Unknown
COD	2,159.43	Unknown	Unknown
Credit Card	459.86	Unknown	USB External Hard Drive
PayPal	2,850.11	Unknown	Unknown
Credit Card	661.74	Unknown	Hair Strengthening Shampoo
Credit Card	216.50	Unknown	Unknown
PayPal	2,933.32	Unknown	Men's Leather Belt
Credit Card	599.65	Unknown	Unknown
Credit Card	2,030.57	Unknown	Unknown
PayPal	132.09	Unknown	USB-C Fast Charger
Credit Card	1,535.68	Unknown	Unknown
PayPal	373.56	Joshua Ryan	Men's Leather Belt
COD	288.98	Unknown	Unknown
Credit Card	1,154.89	Unknown	Unknown
Credit Card	1,219.43	Unknown	Hair Strengthening Shampoo
Credit Card	2,148.12	Unknown	Unknown
COD	354.33	Unknown	Beard Grooming Kit
PayPal	1,409.99	Unknown	Men's Leather Belt
PayPal	266.65	Unknown	Ceramic Kitchen Bowl Set
COD	2,764.80	Unknown	Modern Wall Clock
Credit Card	1,105.34	Unknown	Unknown

COD	1,625.38	Unknown	Wireless Bluetooth Speaker
PayPal	945.22	Unknown	Men's Formal Shirt
PayPal	1,537.45	Unknown	Fabric Sofa Throw Blanket
PayPal	1,864.32	Unknown	Modern Wall Clock
COD	322.35	Unknown	USB External Hard Drive
PayPal	1,050.50	Unknown	Unknown
PayPal	393.57	Unknown	Unknown
PayPal	116.77	Melissa Howard	Stainless Steel Cutlery Set
Credit Card	372.33	Unknown	Adjustable Jump Rope
COD	707.77	Unknown	Multi-Purpose Storage Box
PayPal	190.84	Unknown	Unknown
Credit Card	169.99	Unknown	Unknown
Credit Card	486.06	Unknown	Unknown
COD	783.30	Unknown	Outdoor Sports Water Bottle
Credit Card	1,592.49	Unknown	Noise-Canceling Headphones
COD	783.59	Unknown	Men's Roll-On Perfume
PayPal	1,567.36	Unknown	Gaming Mechanical Keyboard
Credit Card	979.43	Unknown	Makeup Brush Set
PayPal	2,552.30	Unknown	4K Home Projector
PayPal	767.17	Unknown	Unknown
COD	180.94	Unknown	Basketball
PayPal	307.43	Unknown	Knee Support Brace
Credit Card	412.67	Unknown	Unknown
Credit Card	230.60	Unknown	Unknown
Credit Card	1,425.78	Unknown	Unknown
Credit Card	1,301.99	Unknown	Unknown
Credit Card	2,670.74	Unknown	Unknown
PayPal	791.34	Unknown	Hair Strengthening Shampoo
Credit Card	204.93	Unknown	Unknown
Credit Card	1,706.17	Unknown	USB-C Fast Charger
COD	199.24	Unknown	Unknown
Credit Card	1,243.04	Unknown	Unknown
Credit Card	503.98	Unknown	Sports Water Hydration Pack
Credit Card	968.19	Christopher Carter	High-Grip Football
COD	166.48	Unknown	Outdoor Sports Water Bottle
Credit Card	2,156.55	Rachel Myers	Unknown
Credit Card	1,029.47	Michael Torres	Unknown
COD	502.11	Daniel Boyd	Unknown
Credit Card	1,512.92	Unknown	Unknown
Credit Card	289.99	Unknown	Multi-Purpose Storage Box
COD	415.73	Unknown	Unknown

Category	Store_Type
Unknown	Outlet
Unknown	Flagship
Beauty	Online
Unknown	Online
Unknown	Outlet
Electronics	Online
Unknown	Online
Unknown	Online
Unknown	Outlet
Home	Flagship
Sports	Outlet
Electronics	Flagship
Unknown	Online
Electronics	Flagship
Sports	Flagship
Clothing	Flagship
Unknown	Online
Home	Online
Electronics	Online
Unknown	Flagship
Unknown	Flagship
Electronics	Flagship
Unknown	Online
Beauty	Online
Unknown	Online
Clothing	Outlet
Unknown	Online
Unknown	Online
Electronics	Outlet
Unknown	Outlet
Clothing	Online
Unknown	Online
Unknown	Online
Beauty	Online
Unknown	Flagship
Beauty	Outlet
Clothing	Online
Home	Flagship
Home	Flagship
Unknown	Outlet

Electronics	Online
Clothing	Online
Home	Flagship
Home	Flagship
Electronics	Online
Unknown	Online
Unknown	Flagship
Home	Online
Sports	Flagship
Home	Flagship
Unknown	Flagship
Unknown	Online
Unknown	Online
Sports	Online
Electronics	Online
Beauty	Online
Electronics	Online
Beauty	Outlet
Electronics	Online
Unknown	Online
Sports	Online
Sports	Online
Unknown	Flagship
Unknown	Online
Unknown	Online
Unknown	Flagship
Unknown	Outlet
Beauty	Outlet
Unknown	Online
Electronics	Flagship
Unknown	Outlet
Unknown	Flagship
Sports	Outlet
Sports	Online
Sports	Online
Unknown	Online
Unknown	Online
Unknown	Flagship
Unknown	Online
Home	Outlet
Unknown	Online

EXCEL FORMULAS & FUNC

SUM FU	
Formula	**Description**
Total Revenue	Sum of all Total_Amount
Total Qty Sold	Sum of all Quantity

AVERAGE	
Formula	**Description**
Avg Unit Price	Mean price across sales

IF / COUNT	
Formula	**Description**
Credit Card Sales	Count where Payment='Credit Card
PayPal Sales	Count where Payment='PayPal'
COD Sales	Count where Payment='COD'
High-Value Sales	Count sales > $1,000

MAX / MIN	
Formula	**Description**
Max Sale Amount	Largest single transaction
Min Sale Amount	Smallest single transaction

TEXT F	
Formula	**Description**
Order Year	Extract year from date
Order Month	Extract month name

DATA QUA	
Formula	**Description**
Missing Quantity	COUNTBLANK on Quantity col
Missing Unit Price	COUNTBLANK on Price col
Duplicate Sale IDs	COUNTIF to find duplicates

TIONS — ANALYSIS RESULTS

UNCTIONS

Excel Formula Used	Result
#NAME?	$83,124.15
#NAME?	183

E FUNCTIONS

Excel Formula Used	Result
#NAME?	$520.37

IF FUNCTIONS

Excel Formula Used	Result
#NAME?	37
#NAME?	25
#NAME?	19
#NAME?	35

FUNCTIONS

Excel Formula Used	Result
#NAME?	$2,933.32
#NAME?	$116.77

UNCTIONS

Excel Formula Used	Result
#NAME?	e.g. 2024
#NAME?	e.g. July

LITY CHECKS

Excel Formula Used	Result
#NAME?	5
#NAME?	2
6	0

DESCRIPTIVE STATISTICS — KEY NUMERIC FIELDS

Metric	N	Min	Max	Mean	Median
Customer Age	51	18	64	42.55	45
Unit Price ($)	79	62.48	996.33	520.37	517.13
Total Sale Amount ($)	79	116.77	2,933.32	1,052.20	838.85
Quantity per Sale	76	1	4	2.41	2.00

LOYALTY LEVEL DISTRIBUTION			
Loyalty Level	Count	Percentage	Excel Formula
Bronze	9	17.6%	#NAME?
Bronze (Imputed)	4	7.8%	#NAME?
Gold	12	23.5%	#NAME?
Platinum	16	31.4%	#NAME?
Silver	10	19.6%	#NAME?

GENDER DISTRIBUTION			
Gender	Count	% of Total	Formula
Male	18	35.3%	#NAME?
Female	33	64.7%	#NAME?

Mode	Std Dev	Range	IQR
55	14.17	46	-1.00
218.04	287.68	933.85	32.79
198.4164	787.72	2,816.55	-271.11
2	1.12	3	-1.00

PIVOT SUMMARY TABLES — MULTI-DIMENSIONAL ANALYS

Pivot 1: Revenue by Payment Type				
Payment Type	# Sales	Total Qty	Total Revenue	Avg Sale
COD	19	39	17,317.26	911.43
Credit Card	37	88	38,953.69	1,052.80
PayPal	25	56	26,853.19	1,074.13
GRAND TOTAL	**81**	**183**	**83,124.15**	**1,026.22**

Pivot 2: Sales by Store Type			
Store Type	# Sales	Total Revenue	Avg Sale
Flagship	23	26,438.59	1,149.50
Online	43	41,847.72	973.20
Outlet	15	14,837.84	989.19

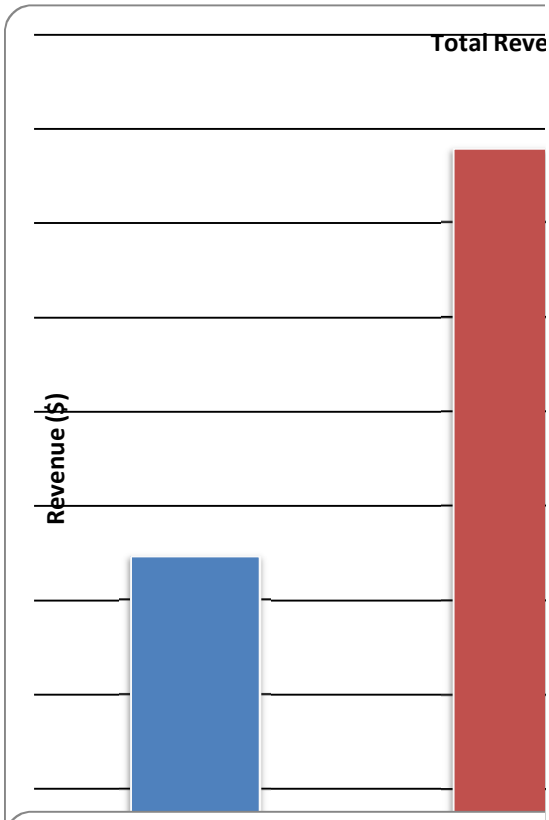
Pivot 3: Sales by Year			
Year	# Sales	Total Revenue	Avg Sale
2023	17	15,815.82	930.34
2024	37	45,616.35	1,232.87
2025	27	21,691.98	803.41

IS

DATA VISUALIZATIONS — CHARTS & ANALYSIS

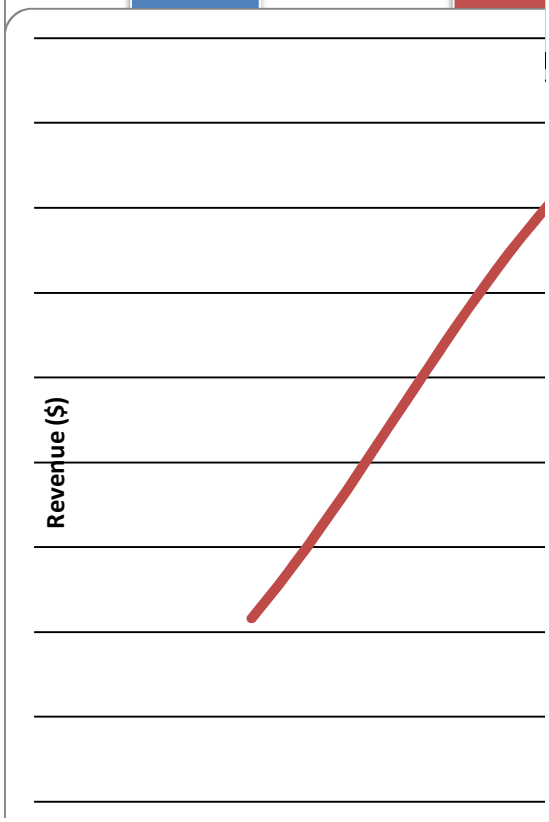
Revenue by Payment Type

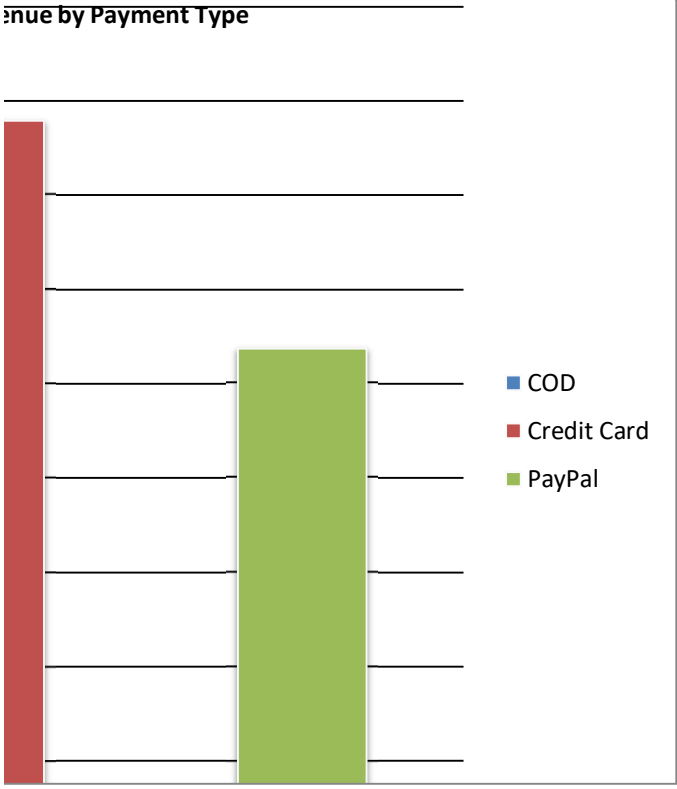
Payment Type	Revenue ($)	# Transactions
COD	17,317.26	19
Credit Card	38,953.69	37
PayPal	26,853.19	25



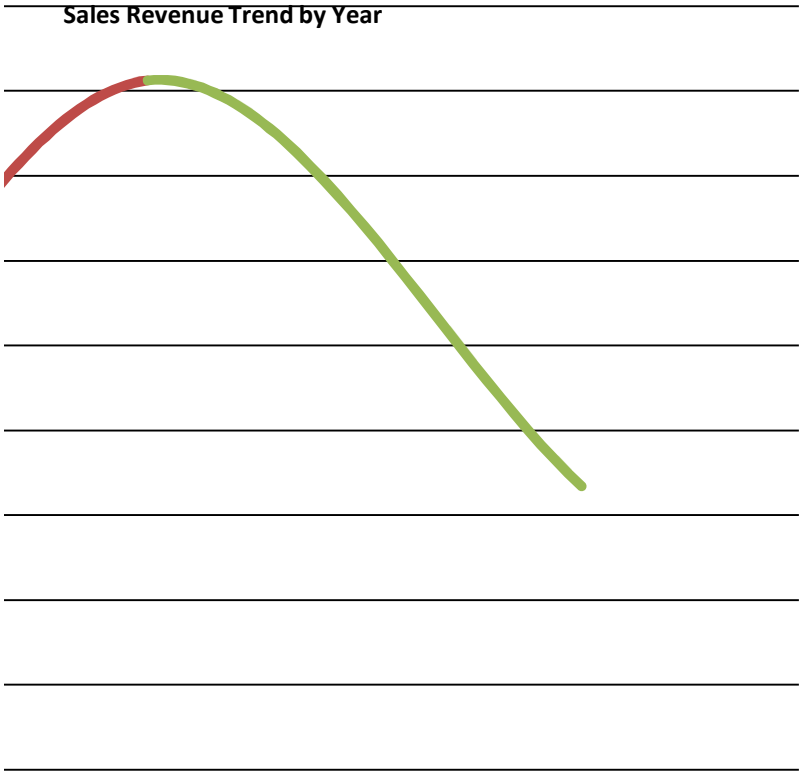
Sales Trend by Year

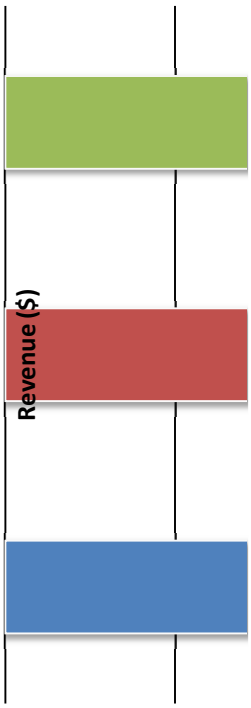
Year	Revenue ($)	# Sales
2023	15,815.82	17
2024	45,616.35	37
2025	21,691.98	27



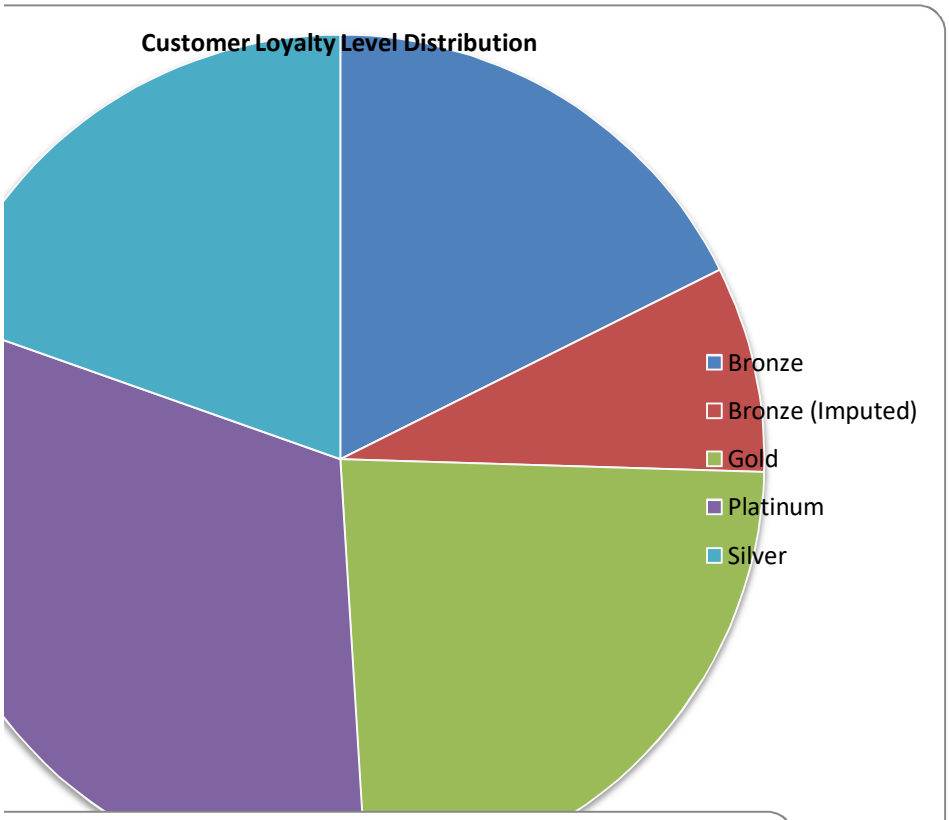

enue by Payment Type
COD
Credit Card
PayPal

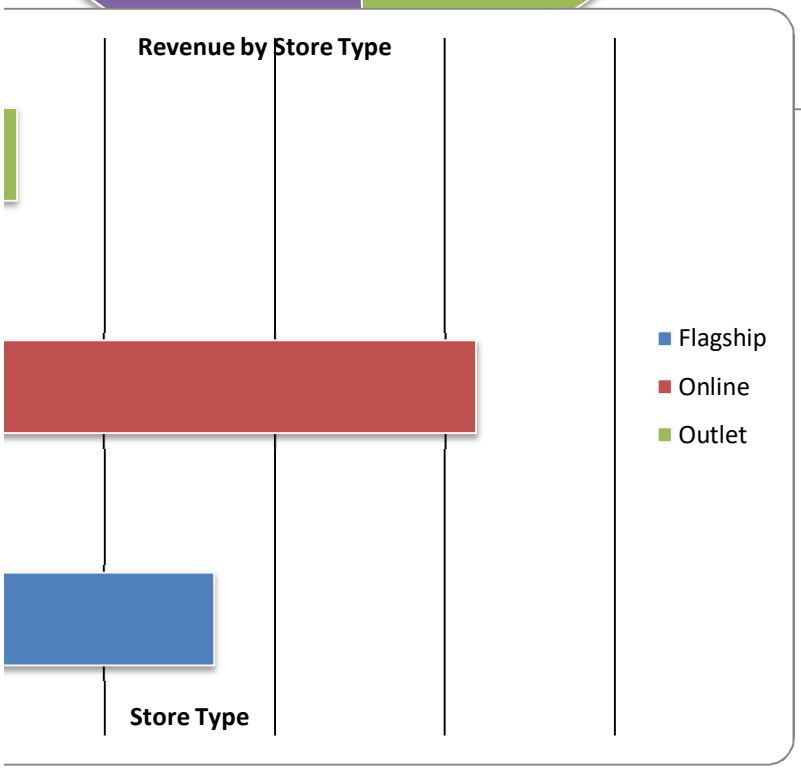
t
9
4
12
16
10


Sales Revenue Trend by Year
2023
2024
2025


Revenue ($)

Year


Customer Loyalty Level Distribution
Bronze
Bronze (Imputed)
Gold
Platinum
Silver


Revenue by Store Type
Flagship
Online
Outlet
Store Type

RETAIL SAL

Data Period: Oct 2023 – Oct 2025 | Samp

Total Revenue	Avg Sale Value
$83,124	$1,026

PAYMENT METHOD BREAKDOWN

Payment Type	Transactions	Revenue	% Share
COD	19	17,317.26	20.8%
Credit Card	37	38,953.69	46.9%
PayPal	25	26,853.19	32.3%

STORE TYPE PERFORMANCE

Store Type	# Sales	Revenue	Avg
Flagship	23	26,438.59	1,149.50
Online	43	41,847.72	973.20
Outlet	15	14,837.84	989.19

CUSTOMER AGE GROUP ANALYSIS

Age Group	Count	%
18-24	6	11.8%
25-34	13	25.5%
35-44	5	9.8%
45-54	15	29.4%
55-64	12	23.5%

LES ANALYTICS DASHBOARD

ple: 81 Sales Records | 500 Customers | 100 Products | 20 Stores

Total Transactions
81

Top Payment
Credit Card

YEAR-OVER-YEAR SALES

Year	Sales Count	Revenue
2023	17	15,815.82
2024	37	45,616.35
2025	27	21,691.98

LOYALTY LEVEL SNAPSHOT

Level	Count	%
Bronze	9	17.6%
Bronze (Imputed)	4	7.8%
Gold	12	23.5%
Platinum	16	31.4%
Silver	10	19.6%

eps documented in sheets 1–6 | Pivot analysis in sheet 9 | Charts in sheet 10